

0.1 RQ3-Specific Coding Codebook

Coding purpose. RQ3 examines which building subsystems the included reviews cover, at what level of control coordination, and under what declared scope. Sixteen units are coded in three groups that are reported separately and never pooled: nine building subsystems, three coordination and interface levels, and four declared-scope dimensions.

What the scale measures. The shared 0–3 scale records treatment intensity, meaning how prominently a review treats a coded unit within its own account of the literature. It does not measure how well reinforcement learning performs on that subsystem, how much primary research exists on it, or how important it is to building operation. A high score indicates that a review gives the unit a prominent and organizing place in its synthesis. The score describes the review, not the field.

Table 1: Shared treatment-intensity scale as applied in RQ3.

Score	Label	General interpretation
0	Not addressed	The unit is not identifiable anywhere in the review, including its figures, tables, keywords, and section headings.
1	Mentioned	The unit is named in a keyword, an enumeration, a single sentence, or one table cell, without explanation of how reinforcement learning applies to it.
2	Discussed	The unit receives paragraph-level treatment, or holds its own row or column in an extraction table that carries entries of its own, or forms its own category in a figure. The review has organized information about the unit rather than merely naming it.
3	Structuring role	The unit organizes part of the review, through a dedicated subsection, a standalone table or figure devoted to it, a taxonomy category, an explicit research question, or a named future-work theme.

Coding units. Group A covers nine subsystems: HVAC; domestic hot water; lighting; appliances; electric vehicles; battery and electrical storage; photovoltaics and on-site renewables; indoor air quality and ventilation; and windows and blinds. Group B covers three coordination levels and one external interface: whole-building HEMS and BEMS; grid, demand-response and microgrid interaction; and multi-zone, district or cluster coordination. Group C covers four declared-scope dimensions: broad scope, HVAC scope, multi-subsystem scope, and deployment scope. Group A is charted; Group B is tabulated; Group C populates the scope heatmap.

Unit boundaries. Several units are adjacent, and consistency depends on routing evidence to one of them rather than both. Table 2 records the routing rules applied.

Zero and blank. Two non-positive outcomes are distinguished and are not interchangeable.

A score of 0 means the review does not address the unit. The scale does not separate a unit that a review considered and did not develop from one that fell outside the review’s declared scope; both are recorded as 0. This is a deliberate simplification, consistent with the construct, which records the treatment a review gives a unit rather than the reasons for that treatment. A score of 0 should therefore not be read as evidence that a review overlooked a subsystem it could have covered. The review’s own declared scope is recorded separately in Group C.

A blank means the unit had no column in that review’s audit table and was never read for. Blanks occur only in Group B, in six of the twenty-two reviews, because those three units lie outside the subsystem analysis and were not re-coded during revision. Blanks were not back-filled with zeros. Counts over Group B therefore state an explicit denominator of 19, 17 and 17

Table 2: Routing rules for adjacent coding units. Evidence is assigned to one unit only; the rules were applied uniformly across the corpus.

Adjacent units	Routing rule
HVAC / IAQ-ventilation	Ventilation is coded as HVAC when it is treated as an actuator serving thermal comfort, and as IAQ-ventilation when air quality is the stated control objective or constraint.
Lighting / windows-blinds	Artificial illuminance control is lighting; daylight admitted or blocked through glazing and shading is windows-blinds.
Appliances / EV / DHW	Electric-vehicle charging is EV, hot-water demand is DHW, and appliances covers the remaining schedulable domestic loads.
Battery storage / EV	Stationary battery systems are battery storage; vehicle batteries are EV, even when used for vehicle-to-building supply.
PV-renewables / grid interface	On-site generation assets are PV-renewables; grid-supplied renewable energy with no on-site asset is the grid interface in Group B.
Group A / Group B	What is controlled is Group A; the level at which control is coordinated, and the external signal it responds to, is Group B.
Group A / Group C	Coverage of a subsystem is Group A evidence. Group C records only what the review declares its scope to be, in its title, abstract, stated objectives, or eligibility criteria.
Operational control / design	Envelope design, insulation, sizing, placement, and retrofit decisions are outside RQ3, which codes operational control only.

rather than 22. The Group A matrix contains no blanks: all 198 cells are coded, and the analysis script asserts this at run time and fails rather than plotting an incomplete matrix.

Evidence rules. Tables, figures, taxonomies, and dedicated sections are coding evidence, and organizing use of a unit in any of them supports a structuring score. A unit holding its own row or column in an extraction table, with entries of its own, satisfies the criterion for 2 because the review has organized information about it. Repeated mentions do not accumulate into a higher score, and paragraph-level discussion, however extensive, is scored 2: length is not structure. Content appearing in the abstract or keywords alone is scored at most 1. A unit appearing only inside the title of a cited work in the reference list is not coded as coverage, since that reflects the cited paper’s subject rather than the review’s own treatment. Where evidence falls between two adjacent levels, the lower score is recorded unless the higher criterion is clearly satisfied, so score-2 and score-3 counts are lower bounds.

Worked boundary examples. The examples below are taken from the survey-level audit tables that follow.

Table 3: Worked boundary examples for RQ3 coding.

Review	Unit	Score	Basis for the boundary
Han et al. (2021)	IAQ–ventilation	2	Air quality is explained as an objective entering the control problem alongside thermal comfort, not merely named. Boundary against 1.
Weinberg et al. (2023)	IAQ–ventilation	2	Air quality is treated as a distinct controlled quantity in the review’s own discussion of building energy systems. Boundary against 1.
Sierla et al. (2022)	Windows–blinds	2	Window and damper actuation is explained as an action type within the review’s classification of actuator abstraction levels. Boundary against 1.
Kannari et al. (2025)	Windows–blinds	2	Window operation appears as a control action in the review’s account of real building deployments, with entries of its own. Boundary against 1.
Latoń et al. (2024)	HVAC	1	HVAC is named among controllable loads in a home energy management review whose development is directed at scheduling and storage. Named, not explained. Boundary against 2.
Pinthurat et al. (2024)	Battery storage	3	Energy storage is the review’s declared organizing subject and structures its tables and synthesis.
Li et al. (2025)	PV–renewables	3	Building-integrated photovoltaics is the review’s organizing category throughout.
Al Sayed et al. (2024)	Lighting, appliances, EV	0	An HVAC-scoped review in which these subsystems are not identifiable. Scored on the not-addressed criterion, not as an omission.
Kannari et al. (2025)	Grid interface (Group B)	blank	The unit had no column in this review’s audit table and was not read for. Recorded blank, not 0, and excluded from the Group B denominator.
Mazhar et al. (2026)	Broad scope (Group C)	3	Title, objectives and coverage all declare a general smart-building energy scope rather than a named subsystem.
Chatterjee and Khovalyg (2023)	Multi-subsystem scope	0	The review declares a single-subsystem scope, the dynamic indoor thermal environment. Group C records the declaration; its HVAC coverage is recorded in Group A.